MINDBURNER MIDI PRODUCTS

NIBBLE

Chiptune Synthesiser, Drum Kit & Sequencer



User Manual | Firmware v0.9.6

Hand-built in Belfast, Northern Ireland

Welcome

NIBBLE brings the sound of 1980s games consoles and home computers to your MIDI setup. It recreates the building blocks of the classic sound chips (pulse waves, the stepped NES triangle, noise and 4-bit wavetables) and adds the tricks chip musicians relied on: fast arpeggios, stepped volume envelopes, pitch sweeps, duty-cycle modulation and tracker-style macros.

Alongside the **8-voice synthesiser** you get a **wave editor** to draw your own 4-bit waves, a live **oscilloscope**, tracker-style **fake echo**, a **chip drum kit with 1-bit speech** on its own MIDI channel, a **song sequencer** with drums, lead and bass tracks, a built-in demo tune with a dancing spider, and 40 patch memories. Everything is synthesised in real time by firmware written from scratch for this instrument.

Connections

Socket	Details
DC power	9 V DC (9 to 12 V accepted), 500 mA or more, 5.5 x 2.1 mm plug. Either polarity is fine. A good regulated adaptor gives the quietest sound.
USB-C power	5 V from a phone charger or power bank. A computer port also works but can add some background noise; it is mainly for firmware updates. USB-C lead not supplied.
MIDI IN	Standard 5-pin DIN. Plays the synth and the drum kit, and receives CC, Program Change, pitch bend, MIDI clock and Start/Stop.
MIDI THRU	Standard 5-pin DIN. Passes on everything arriving at MIDI IN.
Audio out	Stereo line level from a high-quality DAC. The output is hot: start with your mixer gain low.

Start-up

At power-up NIBBLE plays its intro: the Mindburner logo, the artwork, then the NIBBLE title, each letter landing with a chip sound. It then opens on the last patch you used.

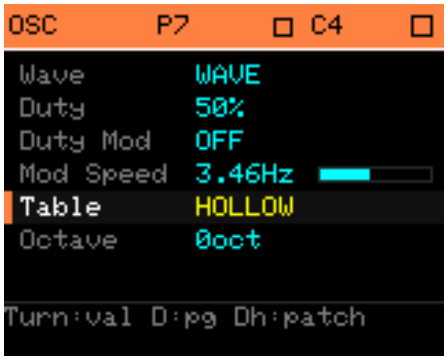
The controls

NIBBLE has four buttons, **A**, **B**, **C** and **D**, and a rotary encoder that also pushes as a button.

Sound pages

Control	What it does
Encoder turn	Change the selected value
Encoder push	Next parameter. Hold: previous page
A tap / A hold	Next parameter / next page (keeps stepping while held)
B / C	Raise / lower the value (hold to repeat and speed up)
D tap / D hold	Next page / open the patch page
B+C hold	Open the patch page (and again to return)
A+B hold 1 s	SEQUENCER screen on / off
A+C hold 1 s	Demo tune on / off
C+D hold 1 s	SCOPE screen on / off
A+B+C hold 2 s	Restore factory patches (asks which first)

Reading the screen

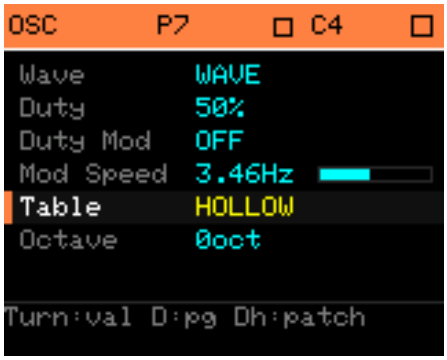


The coloured header shows the **page name**, the **patch number** (with ***** when it has unsaved edits), a MIDI activity square, the **last note** played and a square that fills while a note sounds. Each page has its own colour from the classic console palettes. The selected parameter is highlighted, with its value in yellow. The bottom line names any CC as it arrives and shows messages.

The pages, in order: **OSC, WAVE ED, CHIP, ARP, ENV, MOD, FX, F.ECHO, DRUMS, VOICE, SETUP**. SETUP holds settings for the whole instrument; everything else is stored with each patch.

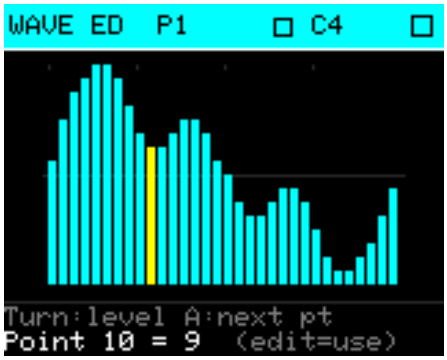
The sound pages

OSC - the oscillator



Parameter	Description
Wave	PULSE, TRI (NES-style 4-bit triangle), SAW (4-bit stepped), NOISE, METAL (short, pitched noise) or WAVE (4-bit wavetables)
Duty	Pulse width: 12.5%, 25%, 50% or 75%
Duty Mod	OFF, CYCLE (steps through the widths, the Game Boy "wah") or PWM (smooth C64-style sweep)
Mod Speed	Speed of the duty modulation
Table	Wavetable for WAVE: SINE, TRI, SAW, SQUARE, HOLLOW, ORGAN, BUZZ, BELL or **USER** (your own, from WAVE ED)
Octave	-2 to +2 octaves

WAVE ED - draw your own wave

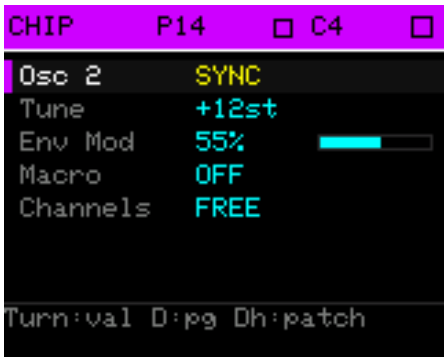


Like the Game Boy wave channel, the shape is 32 steps, each with 16 levels. The selected step is yellow and the bottom line shows its number and level.

Turn the encoder (or press B / C) to set the level. **Tap A or push the encoder** for the next step. Your first change switches the patch to Wave = WAVE and Table = USER, so you hear what you draw, and a short note plays as you edit.

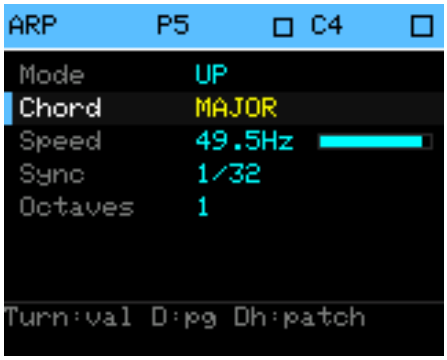
The wave is saved with the patch. Smooth curves sound soft and flute-like; sharp steps and big jumps sound bright and buzzy.

CHIP - chip tricks



Parameter	Description
Osc 2	Second oscillator: OFF, DETUNE (thicker), SYNC (C64 hard-sync lead) or RING (metallic ring modulation)
Tune	Osc 2 interval in semitones (DETUNE: shown in cents)
Env Mod	The envelope sweeps Osc 2 pitch: the classic tearing sync sound
Macro	Tracker-style shape on every note: PLUCK, BOUNCE, WOBBLE, STAB, DROP, TWINKLE, SWEEP, TREMOLO, BWOW, BELL, STRUM, OCT HIT
Channels	FREE (8 voices), NES (2 pulses above middle C, triangle bass below), GB (pulse + wavetable) or C64 (smooth SID-style envelopes)

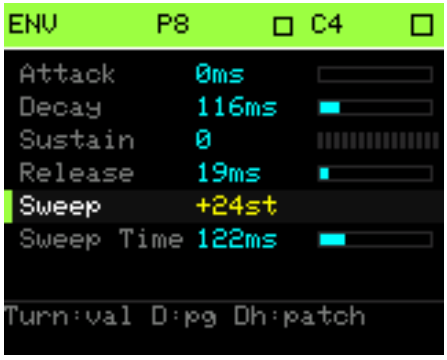
ARP - arpeggiator



One finger plays the whole chord, the classic C64 trick.

Parameter	Description
Mode	OFF, UP, DOWN, UP/DN or RANDOM
Chord	OCTAVE, MAJOR, MINOR, FIFTH, MAJ7, MIN7, DIM, SUS4
Speed	Free-running speed
Sync	FREE, or locked to MIDI clock at 1/8, 1/16 or 1/32 (each with triplets). Falls back to Speed without clock
Octaves	1 to 3

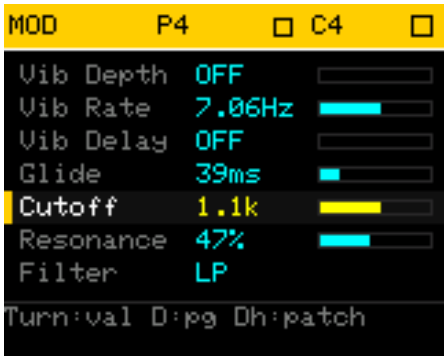
ENV - envelope and sweep



Patch 8 Laser Zap: a big fast sweep.

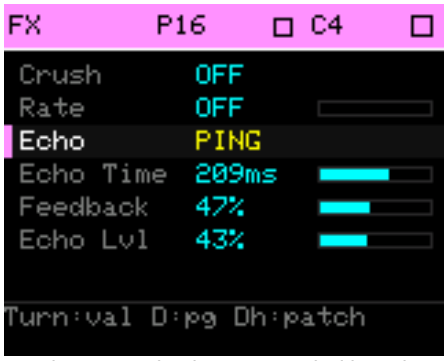
Parameter	Description
Attack / Decay / Release	Envelope times
Sustain	0 to 15, in 16 steps like the original chips
Sweep	Pitch slide into each note in semitones: above 0 falls from higher, below 0 rises from lower
Sweep Time	How long the slide takes

MOD - vibrato, glide and filter



Parameter	Description
Vib Depth / Rate / Delay	Vibrato, with a delay so it fades in as a note is held
Glide	Portamento time
Cutoff / Resonance	C64-style resonant filter (fully up = filter off)
Filter	LP, BP or HP

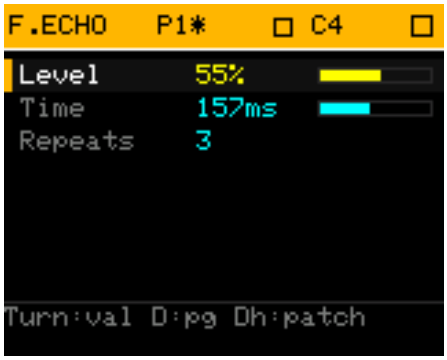
FX - crush and echo



Crush, Rate and Echo are stored with each patch and also affect the drums and the demo tune.

Parameter	Description
Crush	Bit depth from 16 down to 4 bits
Rate	Sample-rate reduction for extra grit
Echo	OFF, STEREO or PING (ping-pong)
Echo Time / Feedback / Echo Lvl	The digital echo

F.ECHO - tracker fake echo

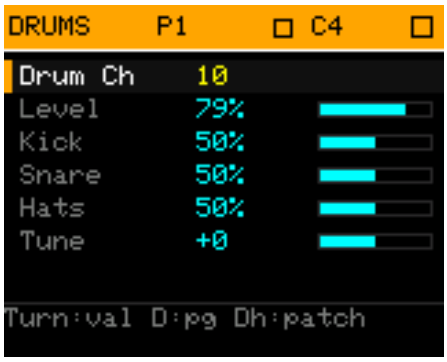


Parameter	Description
Level	How loud the first repeat is (each one is quieter again). OFF turns it off
Time	30 to 600 ms between repeats
Repeats	1 to 3

Works in POLY with Channels FREE or C64.

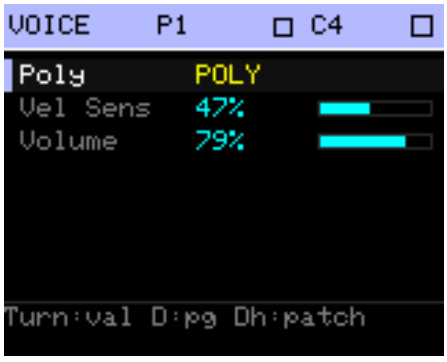
Real chip music had no echo effect, so musicians faked it: each note was played again, quieter, on a spare channel. F.ECHO does exactly that, using free voices. It never cuts off a note you are holding, and sounds more authentically 8-bit than the digital echo on the FX page.

DRUMS - the chip kit



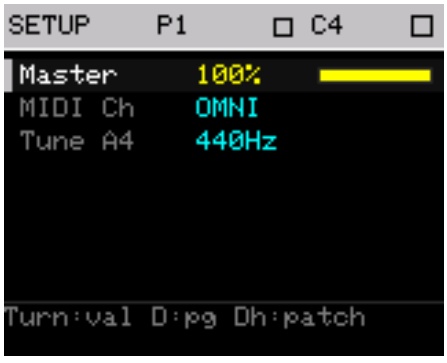
Parameter	Description
Drum Ch	MIDI channel for the kit (default 10, or OFF)
Level	Kit volume (and the speech)
Kick / Snare / Hats	Kick pitch, snare snap, hat length
Tune	Tunes the whole kit, including the speech, up or down an octave

VOICE



Parameter	Description
Poly	POLY (8 voices) or MONO (legato, with glide between held notes)
Vel Sens	Velocity sensitivity
Volume	Patch volume

SETUP - the whole instrument



Parameter	Description
Master	Overall output level, remembered. Recording? Start at about 50%
MIDI Ch	Synth receive channel: OMNI or 1 to 16
Tune A4	430 to 450 Hz master tune, 440 Hz as standard

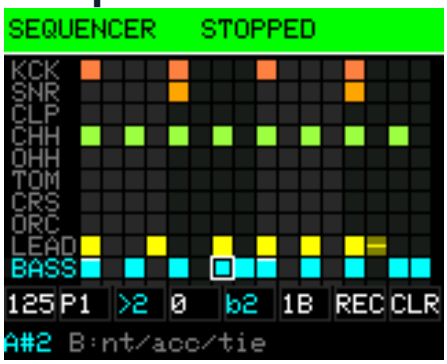
Drum kit and speech

The kit plays on its own MIDI channel (10 by default, the General MIDI drum channel), so you can sequence a whole chip track into one box. The drums run through Crush, Rate and Echo too.

Drum	MIDI notes	Sound
Kick	35, 36	4-bit triangle with a fast pitch drop
Snare	38, 40	Noise plus a short tone
Clap	39	Three quick noise bursts
Closed hat	42, 44	Short metallic noise (chokes the open hat)
Open hat	46	Longer metallic noise
Toms	41, 43, 45, 47, 48, 50	Triangle with a pitch drop, tuned by note
Crash	49, 51, 57	Long noise wash
Zap	37, 56	Pulse sweeping down, the classic game laser
Orchestra hit	55 (52 lower)	Crunchy 4-bit orchestra stab
Speech	60 to 67	1-bit voice: **GO, YEAH, ONE, TWO, THREE, FOUR, READY, GAME OVER**

The speech is stored the way the NES stored its sampled sounds, as 1-bit delta data, so it has that crunchy console character. Drum Tune makes it higher (chipmunk) or lower (robot). Voice made with Flite from Carnegie Mellon University.

Sequencer



Hold **A+B** for one second. NIBBLE has a complete song sequencer: **8 patterns**, each one bar of 16 steps with eight drum rows (KCK, SNR, CLP, CHH, OHH, TOM, CRS, ORC) and two note tracks: **LEAD** (yellow), played by the current patch, and **BASS** (cyan), played by its own patch like a separate chip channel. A thin strip under the grid shows the loop recorder. The sequencer keeps playing when you leave the screen, and everything is saved automatically.

Moving around

Control	What it does
A tap / encoder turn	Next step (on a note step the encoder sets the pitch instead)
A hold / D tap	Next row
Encoder push	Jump between the grid and the boxes
Encoder push hold	MIX screen (see below)
B (on a drum step)	Hit on / off
B (on a LEAD / BASS step)	Rest > note > ACCENT > TIE > rest
C / D hold	Play / stop

On a LEAD or BASS step, **turn the encoder** to set the pitch (C1 to C7); the bottom line shows the note name, with **!** for an accent. An accent plays louder (white bar on top of the step). A **tie** holds the previous note through that step, for long notes and 303-style slides. New bass notes start an octave lower. The step being edited has a white box around it. B and C act when released, so holding A+B to leave never changes a step.

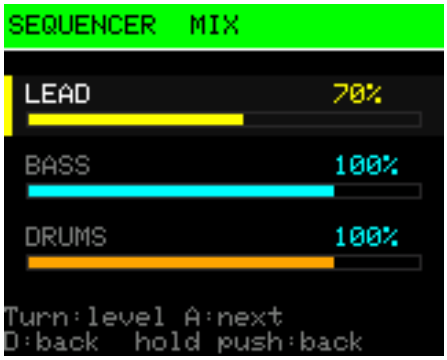
The boxes

Box	B / C (or the encoder)
BPM	Tempo, 60 to 200, with the encoder only (follows MIDI clock when it arrives)
PAT	Pattern 1 to 8 to edit. While playing it switches at the end of the bar
CHN	Chain: OFF, or >2 to >8 plays patterns 1 to N in turn as a song (empty ones skipped). The screen follows the song; MIDI Start begins at the top
KEY	Transpose both note tracks by up to 12 semitones, live
BP	The BASS part's patch: OFF or 1 to 40 (default 2, NES Bass). It uses voices 7 and 8 while the sequencer plays
BAR	Loop recorder length: 1, 2 or 4 bars
REC	B: arm / disarm recording. C: play / stop
CLR	B: clear the recorded loop. C: clear the current row of this pattern

Recording a loop

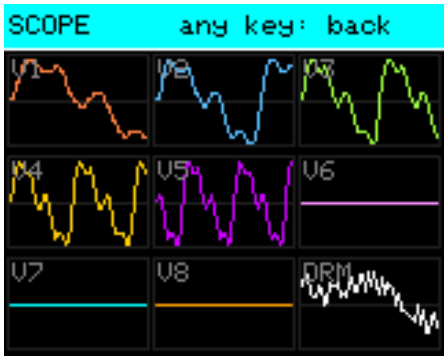
Play the sequencer, select REC and press B to arm it (the box turns red). Play on the synth channel: notes snap to the nearest 16th, with their lengths, and play back on the next pass, up to four at once. The loop runs alongside the patterns at its own length (BAR). Use CLR and B to start again.

MIX



In the sequencer, **hold the encoder push** to open the MIX screen: levels for **LEAD**, **BASS** and **DRUMS**. Turn the encoder (or B / C) to set a level, tap A or push for the next one, and tap D (or hold the push) to go back. 100% is the sound as the patch makes it; you can go from 0 to 127%. The mix is saved with the sequencer and does not change your patches.

Scope



Hold **C+D** for one second to open a live oscilloscope: one window for each of the 8 voices and one for the drums, each in its own colour. Watch the pulse widths change, the arpeggios flicker and your drawn waves come to life. Any key goes back.

Demo tune

Hold **A+C** for one second to play the built-in demo: an original 8-bar chip track at 140 BPM with lead, bass, arpeggio and drums, while the NIBBLE spider dances on a disco floor. You can play along on top. Hold A+C again to stop.

Patch memory



Hold **D** (or B+C) to open the patch page: 40 slots in an 8 x 5 grid, with the live patch in green. Turn the encoder or tap A to choose a slot, and **push the encoder to hear it** without losing your sound; hold the push to go back to yours. Press **B** to load or **C** to save, then confirm with B (yes) or C (no). Tap D to return. Program Change 0 to 39 also selects patches 1 to 40.

Edits are not kept until you save them to a slot.

Restoring factory patches

Hold A+B+C for two seconds. Press B to restore the selected slot, A to restore all 40, or C to cancel.

Factory patches

#	Patch	#	Patch	#	Patch	#	Patch
1	NES Lead	11	Explosion	21	Jump	31	Basic Pulse 50%
2	NES Bass	12	Metal Pad	22	Hit	32	Basic Pulse 25%
3	GB Square	13	Organ Wave	23	Chip Brass	33	Basic Pulse 12.5%
4	C64 PWM Bass	14	Sync Lead	24	Wobble Bass	34	Basic Triangle
5	Major Arp	15	Square Bass	25	Bounce Pluck	35	Basic Saw
6	Minor Arp	16	Echo Lead	26	Twinkle	36	Basic Noise
7	GB Wave	17	Chord Stab	27	Drop Bass	37	Basic Metal
8	Laser Zap	18	Lo-Fi Pad	28	Sweep Pad	38	Basic Wave
9	Power Up	19	1-Bit Beeper	29	Tremolo Organ	39	Basic Bass
10	Coin	20	Ring Bell	30	Strum	40	Basic Arp

Patches 31 to 40 are plain starting points with no effects. Some factory sounds use Crush, Rate or Echo on purpose (for example 11 Explosion and 19 1-Bit Beeper): if a sound is gritty or echoey, check the FX page.

MIDI implementation

Page	CC numbers
General	1 Mod wheel (vibrato), 5 Glide, 7 Volume, 120/123 All notes off, 126 Mono, 127 Poly
OSC	105 Wave, 106 Duty, 107 Duty Mod, 108 Mod Speed, 109 Table, 110 Octave
CHIP	20 Osc 2, 21 Tune, 22 Env Mod, 23 Macro, 24 Channels
ARP	102 Mode, 103 Chord, 104 Speed, 25 Sync, 111 Octaves
ENV	73 Attack, 75 Decay, 79 Sustain, 72 Release, 112 Sweep, 113 Sweep Time
MOD	77 Vib Depth, 76 Vib Rate, 78 Vib Delay, 74 Cutoff, 71 Resonance, 26 Filter
FX	114 Crush, 115 Rate, 116 Echo, 117 Echo Time, 118 Feedback, 119 Echo Level
F.ECHO	28 Level, 27 Time
DRUMS	85 Level, 86 Kick, 87 Snare, 88 Hats, 89 Tune
VOICE	90 Vel Sens

Also received: Note On/Off, pitch bend (plus or minus 2 semitones), Program Change (patches 1 to 40), MIDI Clock (arpeggiator and sequencer), Start and Stop.

Tips and troubleshooting

If...	Try this
No sound	Check MIDI Ch on the SETUP page is OMNI or matches your keyboard, and that Master is up.
No drums	Check Drum Ch on the DRUMS page matches the channel your controller sends drums on.
Every third note sounds different	Channels is set to NES or GB on the CHIP page. Set it to FREE.
Sound is gritty or echoey	The patch uses Crush, Rate or Echo on the FX page, or F.ECHO. Turn them off.
F.ECHO does nothing	It needs POLY on the VOICE page and Channels FREE or C64.
Zipper noise with several units	Give each unit its **own** power supply, or use a pedal power supply with **isolated** outputs (300 mA or more each). Sharing one adaptor or a daisy-chain lead between units lets their digital noise into each other's audio through the shared ground.
Clicks or crackle on USB	Computer ground noise: use the DC adaptor, a charger or a power bank.
Lead or bass too loud in a song	Hold the encoder push in the sequencer for the MIX screen.
Recording distorts	SETUP > Master to about 50%, recorder gain low.

Mindburner MIDI Products, hand-built in Belfast, Northern Ireland.